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CTO and Founder of Netradyne

the incident and send a report to the fleet manager. We're using this technology, not to take control of the vehicle, but to assist the drivers to reduce the number of accidents.

How does Netradyne work in a commercial fleet vehicle?

The device known as Driveri has cameras that can simultaneously process all the signs, vehicles, and traffic lights in front of the vehicle, as well as keep track of the drivers' activities, to check if they are drowsy or distracted etc. The system uses high-end processors and algorithms to essentially do two things - one is to assess the performance of the drivers and provide real-time alerts through voice messages, and the second is to send this information along with the video for every single alert, to the fleet managers who can then coach the drivers or consider them for incentives, depending upon the drivers' performance. Additionally, every driver starts accumulating a score based on the manner of their driving which will come in handy in the future.

How is a driver's privacy taken care of?

Privacy is an important aspect to us, and we have taken the necessary steps toward ensuring it. When the vehicle is switched off, the inward camera immediately shuts off, this suggests that the device monitors the driver only when the vehicle is on. I do not dispute that there is a fine line between privacy and safety, but we have put in a lot of safeguards

to ensure utmost privacy. Now, numerous drivers themselves recognise how beneficial this device is for them and insist on having it installed in their vehicles.

How are drivers realising that Netradyne's Driveri favours them?

When an accident occurs around a commercial fleet vehicle, 70% of the time, the drivers of these vehicles are blamed. With our device recording every alert and storing it in the cloud, one can use this data to check what happened during the drive time. So, a lot of the drivers have realised that this is advantageous to them and insist on having such a camera on their windshield. Besides, drivers also realise that with a higher driving score, they are eligible for more bonuses. So, they can earn more for safe driving. In fact, there was a driver who told us that he was initially quite apprehensive about a device monitoring him. However, he went on to say that the device noticed him being drowsy and tired from driving for a long period and alerted him through a voice message to wake him up. This is an example of our device's intelligent assessment of the situation to prevent mishaps.

How does the geography of a place affect the accuracy of the device?

The device is installed on vehicles that operate out of regions in freezing, subzero temperatures, and in the hottest parts of the Earth. The system must be designed to factor that in. A lot of effort goes into system reliability and performance in all these harsh conditions and the answer is because of all the efforts there is no impact on accuracy. It is designed for a wide range of environmental conditions, and it has to be designed that way.

What is the major differentiator between Netradyne and its competitors?

We are the category-defining leader in this aspect because we practically started the demand for a space like this that focuses on prevention rather than exoneration. Of course, many companies began installing cameras on the windshield of commercial vehicles 10-15 years ago. However, they were mostly used for exoneration. We, on the other hand, have devised a way to prevent accidents - an algorithm that is extremely similar to what we're using for autonomous driving. We are a pioneer, in terms of alerting the driver, keeping a driving score, and sending this information to the manager. As is the

case with any successful company, others have tried to catch up. However, we are far ahead of others in terms of technology. Above all, we are manufacturing this high performing device in India, for the world.

Netradyne recently received funding of \$150 million in its Series C funding. How do you see that?

Artificial Intelligence is all about machine learning. The power of the system is that the more data you get, the more you can learn. We have analysed 4 billion miles of driving and a billion minutes of analysis every month. This gives us a vast amount of data to persistently improve and perfect our systems. I think the funding is reflective of that. The market is huge because it affects all commercial vehicles and is also directly dramatically affecting commercial auto insurance. We have data from our customers that suggest that the accident rates have reduced to 30-40%, and distracted driving has gone down by about 80%, which is all a very meaningful reduction in accidents and compliance. I think this is why SoftBank invested \$150 million in us.

What does the future look like for Netradyne?

I think there continues to be a huge opportunity to make the system more sophisticated and cover a lot more aspects. New features that we keep adding are to not only prevent significant accidents but also fender benders. These require advanced algorithms because the device must be able to separate these minor accidents from potholes, speed breakers etc. But that apart, geographically too, we are planning to launch in Europe, which will be happening soon and eventually will be expanding in APAC. We want to become a global player and you'll see continued expansion in that place.

So how is an advanced vision-based driver recognition and fleet safety solution built? What keeps Driveri well ahead of other products in the industry? The company's experts that have toiled relentlessly since the day it was conceptualized shed light on the technology, cloud services and analytics behind their product that is accurate 99% of the time. At Netradyne, Driveri has been developed by a world-class team of scientists in India, for the world. Scientists and engineers who have unmatched expertise in the field of computer vision, deep learning, edge computing, and predictive analytics. We also caught up with Teja Gudena, Senior Vice President, Engineering, Devices at Netradyne, who